E-Commerce Order Management System

Week 3

Task III: Relational Schema

Introduction

The relational schema is derived from the Entity Relationship (ER) Diagram created for the E-Commerce Order Management System. It defines the database tables along with their attributes, primary keys, and foreign keys that will be implemented in the database.

Relational Schema

CUSTOMER

Attribute	Description
CustomerID (PK)	Unique Customer ID
Name	Customer Name
Email	Customer Email
MobileNumber	Contact Number
Address	Customer Address
RegistrationDate	Registration Date

CATEGORY

Attribute	Description
CategoryID (PK)	Unique Category ID
CategoryName	Category Name
Description	Category Description

SUPPLIER

Attribute	Description
SupplierID (PK)	Unique Supplier ID
SupplierName	Supplier Name
ContactInformation	Contact Details

PRODUCT

Attribute	Description
ProductID (PK)	Unique Product ID
Price	Product Price
StockQuantity	Available Stock
CategoryID (FK)	References Category
SupplierID (FK)	References Supplier

ORDER

Attribute	Description
OrderID (PK)	Unique Order ID
CustomerID (FK)	References Customer
OrderDate	Order Date
TotalAmount	Total Amount
OrderStatus	Order Status

ORDER_DETAILS

Attribute	Description
OrderDetailID (PK)	Unique Order Detail ID
OrderID (FK)	References Order

ProductID (FK)	References Product
Quantity	Quantity Ordered
UnitPrice	Product Price

PAYMENT

Attribute	Description
PaymentID (PK)	Unique Payment ID
OrderID (FK)	References Order
PaymentMethod	Payment Method
PaymentDate	Payment Date
PaymentStatus	Payment Status

SHIPMENT

Attribute	Description
ShipmentID (PK)	Unique Shipment ID
OrderID (FK)	References Order
DeliveryAddress	Delivery Address
ShipmentDate	Shipment Date
DeliveryStatus	Delivery Status

REVIEW

Attribute	Description
ReviewID (PK)	Unique Review ID
CustomerID (FK)	References Customer
ProductID (FK)	References Product
Rating	Product Rating
Comments	Customer Review

Final Relational Schema

```
CUSTOMER(
    CustomerID PK,
    Name,
    Email,
    MobileNumber,
    Address,
    RegistrationDate
)

CATEGORY(
    CategoryID PK,
    CategoryName,
    Description
)

SUPPLIER(
    SupplierID PK,
    SupplierName,
    ContactInformation
)

PRODUCT(
    ProductID PK,
    Price,
    StockQuantity,
    CategoryID FK,
    SupplierID FK
)

ORDER(
    OrderID PK,
    CustomerID FK,
    OrderDate,
    TotalAmount,
    OrderStatus
)
```

```
ORDER_DETAILS(
    OrderDetailID PK,
    OrderID FK,
    ProductID FK,
    Quantity,
    UnitPrice
)

PAYMENT(
    PaymentID PK,
    OrderID FK,
    PaymentMethod,
    PaymentDate,
    PaymentStatus
)

SHIPMENT(
    ShipmentID PK,
    OrderID FK,
    DeliveryAddress,
    ShipmentDate,
    DeliveryStatus
)

REVIEW(
    ReviewID PK,
    CustomerID FK,
    ProductID FK,
    Rating,
    Comments
)
```

Conclusion

The ER Diagram has been successfully converted into a relational schema. Each entity is represented as a separate relation with clearly identified primary keys and foreign keys, ensuring proper relationships and data integrity for the E-Commerce Order Management System.